
Mapping Environmental Pollution and Assessing Stakeholder Awareness at Somapura Mahavihara UNESCO World Heritage Site: A Policy Implementation Study

Abstract

Environmental stress at archaeological heritage sites is increasingly shaped by the interaction of physical deterioration, visitor behavior, local practices, and institutional capacity. Somapura Mahavihara, inscribed on the UNESCO World Heritage List as the Ruins of the Buddhist Vihara at Paharpur, provides a significant case for examining these interactions because its exposed brick architecture, open archaeological landscape, and public accessibility make it vulnerable to multiple forms of environmental pressure while also requiring sustained conservation governance (UNESCO World Heritage Centre, n.d.). This study develops an integrated framework for mapping environmental pollution, assessing stakeholder awareness, and evaluating policy implementation at the site. It proceeds from the argument that heritage degradation cannot be understood solely as a material problem; rather, it must be examined through the linked dimensions of environmental condition, stakeholder practice, and institutional response.

The study adopts a convergent mixed-method case study design. Spatial field observation is used to identify and map visible indicators of environmental stress, including solid waste, plastic litter, stagnant water, drainage-related vulnerability, vegetation intrusion, soil erosion, and visitor-pressure zones. These observations are combined with questionnaire-based assessment of awareness, attitudes, and practices among visitors, local residents, vendors, site personnel, and relevant officials. Semi-structured interviews and document review are further employed to evaluate how conservation and environmental management policies are interpreted and implemented in everyday site practice. Quantitative data are designed to be analyzed through descriptive and inferential statistics, while qualitative findings are interpreted through thematic analysis and triangulated with spatial evidence.

The paper contributes in three ways. First, it advances heritage research by linking archaeological conservation with environmental governance and policy implementation. Second, it offers a methodological model for integrating pollution mapping with stakeholder analysis in an open heritage landscape. Third, it provides a practical basis for strengthening conservation planning, public awareness, and site-level environmental management at Somapura Mahavihara and other archaeological World Heritage properties facing similar pressures. By foregrounding the relationship between environmental risk and institutional practice, the study argues for a more integrated and participatory approach to heritage protection.

Keywords: Somapura Mahavihara; Paharpur; environmental pollution; stakeholder awareness; UNESCO World Heritage; heritage management; policy implementation; archaeological landscape

1. Introduction

1.1 Background

Cultural heritage sites are increasingly exposed to environmental pressures that extend beyond conventional questions of monument preservation. In archaeological landscapes, deterioration often occurs through cumulative processes such as unmanaged waste, water stagnation, erosion, vegetation growth, visitor pressure, and weak environmental monitoring. These pressures are especially consequential where heritage remains are materially fragile, spatially dispersed, and closely tied to landscape setting. Conservation scholarship has therefore emphasized that the protection of world heritage requires not only technical treatment of structures but also the comprehensive management of environmental conditions, human use, and institutional responsibility (Feilden & Jokilehto, 1998; UNESCO, ICCROM, ICOMOS, & IUCN, 2013).

Somapura Mahavihara, internationally recognized through the UNESCO property “Ruins of the Buddhist Vihara at Paharpur,” is one of South Asia’s most important Buddhist archaeological sites and a major monastic complex of the early medieval period (UNESCO World Heritage Centre, n.d.). Its monumental central shrine, monastic enclosure, exposed brick remains, surrounding open grounds, and associated archaeological features collectively form a heritage landscape rather than a single monument. This condition is central to the site’s value, but it also increases vulnerability. Environmental degradation at such a site may affect not only the physical fabric of the monument but also its authenticity, integrity, visual setting, interpretive legibility, and long-term conservation capacity (Labadi, 2013).

The significance of environmental management at heritage sites has grown alongside broader shifts in heritage theory and practice. Heritage scholars have increasingly shown that conservation is not simply a technical activity but a socially managed and politically mediated process shaped by values, actors, and institutions (Smith, 2006). In this regard, environmental pollution at a World Heritage Site should not be understood only as physical contamination or visual disorder. It is also a governance issue connected to awareness, compliance, monitoring, inter-agency coordination, and the conduct of site users. This perspective is particularly relevant where conservation authorities, visitors, local communities, tourism actors, and public agencies all influence site conditions.

1.2 Problem Statement

Although conservation policies often articulate strong commitments to heritage protection, the translation of these commitments into day-to-day management remains uneven. Policy implementation studies have repeatedly shown that policy success depends not only on formal rules but also on administrative coordination, institutional capacity, local compliance, and the behavior of multiple actors operating in real-world settings (Hill & Hupe, 2002; Pressman & Wildavsky, 1973; Sabatier & Mazmanian, 1980). This insight is highly relevant to heritage sites, where environmental degradation may persist even when conservation principles are formally acknowledged.

At Somapura Mahavihara, the coexistence of exposed archaeological remains, open visitor circulation, seasonal hydrological variability, tourism-related use, and broader local activity creates conditions in which environmental stress can accumulate gradually and unevenly across the site. Yet research and management discussions often privilege monument-centered conservation over integrated environmental assessment. When pollution or ecological stress is observed, it is frequently treated as a maintenance issue rather than part of a wider governance and awareness problem. This creates a significant analytical gap. Without mapping the spatial distribution of environmental pressures, understanding how stakeholders perceive and respond to those pressures, and evaluating whether policy is effectively implemented, heritage management risks remaining reactive rather than preventive.

1.3 Aim and Objectives

The aim of this study is to map environmental pollution, assess stakeholder awareness, and evaluate policy implementation gaps at Somapura Mahavihara in order to develop an integrated framework for sustainable heritage management.

The specific objectives are:

1. To identify the principal forms of environmental pollution and environmental stress affecting the site and its surrounding landscape.
2. To map the spatial distribution of these pressures in order to identify vulnerable and high-risk zones.
3. To assess levels of awareness, attitudes, and self-reported practices among key stakeholder groups.
4. To analyze the extent to which conservation and environmental management policies are implemented in everyday site governance.
5. To propose practical recommendations for improving environmental protection, stakeholder participation, and policy enforcement.

1.4 Research Questions

The study is guided by the following questions:

1. What forms of environmental pollution and environmental stress are present within and around Somapura Mahavihara?
2. Which areas of the site are most vulnerable to pollution-related deterioration?
3. What is the level of awareness among key stakeholders regarding environmental threats to the heritage site?
4. How do stakeholder attitudes and practices influence environmental conditions at the site?
5. What gaps exist between conservation policy provisions and actual implementation?
6. What policy and management strategies can improve environmental stewardship at Somapura Mahavihara?

1.5 Hypotheses

The study tests four interrelated hypotheses.

H1: Lower stakeholder awareness is significantly associated with weaker environmentally responsible behavior at the heritage site.

H2: Areas experiencing higher visitor pressure and lower levels of monitoring exhibit higher levels of visible environmental pollution.

H3: Gaps in policy implementation significantly contribute to unmanaged environmental stress at Somapura Mahavihara.

H4: Stakeholders with greater awareness of the site's heritage value are more likely to support conservation-oriented environmental regulations.

1.6 Significance of the Study

This study is significant for both scholarship and practice. Academically, it contributes to the growing literature connecting heritage conservation to environmental governance, stakeholder participation, and implementation analysis. While previous work has emphasized management planning and public involvement in heritage contexts (Aas et al., 2005; UNESCO et al., 2013), fewer studies integrate these concerns with the spatial mapping of pollution in an archaeological landscape. Methodologically, the paper develops a mixed-method design that combines field-based environmental assessment with social and institutional analysis. Practically, the findings are expected to support site managers and policymakers by identifying pollution hotspots, awareness gaps, and policy bottlenecks that can be addressed through targeted intervention.

2. Literature Review

2.1 Heritage Sites as Environmental Landscapes

Heritage conservation has progressively moved away from a narrow focus on individual monuments toward a broader understanding of landscape, setting, and context. This shift reflects recognition that cultural significance is shaped not only by built fabric but also by spatial relationships, environmental conditions, and patterns of use (Feilden & Jokilehto, 1998; UNESCO et al., 2013). In archaeological sites, this perspective is particularly important because remains are often dispersed, excavated, weather-exposed, and dependent on the surrounding ground surface for interpretation and protection. Environmental degradation at such sites may manifest through a wide range of processes, from waterlogging and vegetation growth to litter accumulation and trampling.

Within UNESCO heritage discourse, authenticity and integrity are not preserved solely through material repair; they also depend on maintaining the conditions that allow a site's values to be sustained over time (Labadi, 2013; UNESCO World Heritage Centre, n.d.). At open archaeological sites, environmental stress can degrade physical fabric, obscure archaeological legibility, alter visual setting, and diminish visitor experience. These pressures are often cumulative rather than catastrophic. Their importance lies in their gradual but continuous effect, which may be overlooked in management systems focused primarily on acute threats.

Somapura Mahavihara exemplifies these concerns. As a large monastic complex with extensive brick remains and open grounds, the site must be understood as a heritage landscape whose conservation requires ongoing environmental management. Pollution in this context is not limited to visible solid waste. It includes stagnant water, inadequate drainage, erosion, vegetation intrusion, and unmanaged movement patterns that may affect both structure and setting. A landscape-based analytical framework is therefore more appropriate than a narrow monument-based approach.

2.2 Environmental Pollution and Heritage Conservation

The conservation literature consistently demonstrates that environmental factors are among the most persistent threats to heritage sites. Although public discussion often reduces pollution to littering, the conservation field treats environmental risk more broadly, encompassing moisture, runoff, biological growth, soil instability, visual contamination, and incompatible uses that weaken the integrity of the heritage environment (Feilden & Jokilehto, 1998; UNESCO et al., 2013). For archaeological sites composed of exposed brick and earthen substrates, the interaction between climate, water movement, vegetation, and human activity can significantly accelerate deterioration.

These threats become especially challenging where monitoring is weak or inconsistent. Because many environmental impacts are dispersed and seasonal, they require routine observation, mapping, and maintenance rather than episodic intervention. Yet heritage management systems do not always integrate environmental audit procedures into standard conservation practice. This creates a gap between preservation principles and operational reality. Physical degradation is often addressed only after visible damage becomes pronounced.

A central implication of this literature is that environmental protection at heritage sites must be preventive, spatially informed, and management-oriented. Pollution mapping is therefore not merely descriptive. It is a tool for identifying where risks are concentrated, how they relate to circulation or land use, and which zones require priority intervention. In the context of Somapura Mahavihara, such mapping can reveal the relationship between open access areas, water-prone zones, peripheral archaeological remains, and site maintenance practices.

2.3 Stakeholder Awareness and Participatory Heritage Management

A second major theme in the literature concerns the role of stakeholders in heritage conservation. Heritage is not protected by institutions alone; it is shaped by the conduct, perceptions, and cooperation of multiple actors. Smith (2006) argues that heritage is socially produced through practices of meaning, authority, and public engagement rather than simply inherited as a fixed object. This insight is important for understanding why stakeholder awareness matters. Where users do not understand the significance or fragility of a site, environmentally harmful behaviors may become normalized even in formally protected settings.

Stakeholder collaboration has been identified as a core requirement of effective heritage management, particularly in publicly accessible and tourism-oriented sites (Aas et al., 2005). Visitors, residents, vendors, guides, site staff, and public officials all influence the conditions under which conservation occurs. Their awareness affects how they dispose of waste, respect movement restrictions, respond to site rules, and support or resist regulatory measures. Awareness is not identical across groups. Officials may possess procedural knowledge but face implementation constraints, while visitors may value the site symbolically yet remain uninformed about practical conservation rules.

The literature thus suggests that stakeholder awareness should be treated as both a social and policy variable. It is social because it shapes behavior and public responsibility. It is policy-related because awareness is affected by interpretation systems, signage, education campaigns, communication strategies, and participatory planning. At a site such as Somapura Mahavihara, assessing awareness is essential for understanding whether existing management tools are generating informed stewardship or merely passive visitation.

2.4 Policy Implementation in Heritage Management

Policy implementation research provides a useful framework for examining why heritage protection often falls short in practice. Classic implementation studies emphasize that policy outcomes are rarely determined by legislation alone. They are mediated by organizational structures, resource availability, inter-agency coordination, administrative discretion, and local compliance (Hill & Hupe, 2002; Pressman & Wildavsky, 1973; Sabatier & Mazmanian, 1980). This insight is highly relevant to heritage management, where responsibilities may be distributed across conservation authorities, tourism bodies, local government, environmental agencies, and site-level staff.

In the heritage field, UNESCO and related international guidance repeatedly stress the need for integrated management systems, monitoring, participation, and long-term planning (UNESCO et al., 2013). However, translating these principles into everyday site practice is complex. Gaps may arise when management plans are insufficiently operationalized, when site staffing is limited, when enforcement is inconsistent, or when environmental responsibilities are not clearly institutionalized. Policy may therefore exist at the level of intention while remaining weak at the level of implementation.

This study adopts policy implementation not as an abstract administrative category but as an empirical lens for examining site governance. It asks whether environmental rules and conservation commitments are visible in the practical management of Somapura Mahavihara. Are monitoring and maintenance routines adequate? Are stakeholder responsibilities communicated effectively? Are institutional mandates coordinated? Are pollution-related threats addressed systematically or incidentally? These questions move beyond formal policy presence to evaluate actual policy performance.

2.5 Research Gap

Existing scholarship provides strong foundations for understanding heritage value, environmental risk, stakeholder participation, and policy implementation. However, three gaps remain. First, relatively few studies integrate these strands in a single analytical framework for an open archaeological landscape. Second, within heritage settings, environmental pollution is often discussed in general terms without being spatially mapped in relation to user behavior and governance conditions. Third, South Asian archaeological World Heritage Sites remain underrepresented in empirical studies that connect environmental management to stakeholder awareness and implementation analysis.

This study addresses these gaps by combining pollution mapping, awareness assessment, and policy implementation review within one case study. Rather than treating environmental degradation as a purely material issue, it conceptualizes site condition as the outcome of interaction among environmental pressures, stakeholder practices, and institutional response.

In doing so, the paper contributes a more integrated model for understanding and managing heritage vulnerability.

3. Methodology

3.1 Research Design

This study employs a **convergent mixed-method case study design**. Mixed methods are appropriate because the research problem spans physical, social, and institutional dimensions that cannot be adequately captured through a single approach (Creswell & Plano Clark, 2018). The case study strategy is suitable because the paper investigates a bounded heritage setting characterized by complex interactions among environment, users, and governance structures (Yin, 2018). Quantitative and qualitative data are collected in parallel, analyzed separately, and then integrated during interpretation.

The design has three interrelated components. The first is a spatial-environmental component that maps visible forms of pollution and environmental stress across the site. The second is a social component that measures stakeholder awareness, attitudes, and reported practices through structured questionnaires. The third is an institutional component that evaluates policy implementation through semi-structured interviews and document review. The logic of integration is straightforward: environmental mapping identifies where problems are concentrated, survey data help explain behavioral and perceptual patterns associated with those problems, and qualitative evidence clarifies the governance conditions under which they persist.

3.2 Study Area

The study area is Somapura Mahavihara, the UNESCO World Heritage property officially listed as the Ruins of the Buddhist Vihara at Paharpur in Bangladesh (UNESCO World Heritage Centre, n.d.). The property is significant for its monumental central shrine, monastic cells, associated archaeological remains, and enclosed site layout. Because the site's heritage value is inseparable from its broader spatial setting, the study area includes not only the central monument but also surrounding ruins, circulation routes, open grounds, drainage-prone areas, peripheral archaeological features, and adjacent activity interfaces that may influence environmental condition.

For field analysis, the site is divided into operational survey zones. These may include the central shrine precinct, the enclosed monastic perimeter, peripheral archaeological clusters, visitor access corridors, drainage and water-prone edges, and transitional boundary areas.

Zoning allows environmental observations to be recorded systematically and compared across the site.

3.3 Data Sources

The study uses both primary and secondary data. Primary data consist of field observation, site mapping, structured questionnaires, semi-structured interviews, and photographic documentation. Secondary data include UNESCO site descriptions, conservation and management guidance, relevant national and local policy documents, institutional reports, and prior academic research on heritage management, stakeholder participation, and policy implementation. The use of multiple data sources supports triangulation and strengthens interpretive rigor (Creswell & Plano Clark, 2018).

3.4 Sampling Strategy

The quantitative component uses **stratified purposive sampling** to ensure representation of major stakeholder categories associated with the site. These categories include visitors, local residents, vendors or service providers, site staff, local administrators, and heritage-related officials. A target sample of approximately 200 to 300 questionnaire respondents is appropriate for generating meaningful descriptive and inferential analysis.

The qualitative component uses purposive sampling for key informant interviews. Interviewees are selected on the basis of institutional role, management experience, or direct engagement with the site. This group may include archaeology officials, conservation personnel, local government representatives, environmental officers, tourism actors, guides, and academic experts. Approximately 12 to 20 interviews are anticipated, or until thematic saturation is reached.

3.5 Data Collection Procedures

3.5.1 Environmental Observation and Pollution Mapping

A structured field observation protocol is used to document visible environmental conditions across all survey zones. Observation categories include solid waste accumulation, plastic litter, stagnant water, drainage obstruction, vegetation intrusion, soil erosion, surface degradation, informal pathways, crowding pressure, and other visible disturbances to the archaeological setting. Each observation point is georeferenced where possible and recorded using a standard severity scale ranging from absence to high intensity. Photographs and field notes supplement the coded observations.

The purpose of this stage is to create a spatial distribution map of environmental stress. If GIS facilities are available, observation points are entered into a geospatial database and visualized through hotspot or zone-based maps. If GIS resources are limited, a manually coded site plan

can be used to produce analytical overlays. In either case, the resulting maps identify areas of higher vulnerability and link them to circulation, drainage, and management patterns.

3.5.2 Questionnaire Survey

The questionnaire instrument is designed to measure awareness, attitudes, and self-reported practices related to environmental protection and heritage conservation. The survey includes five sections: respondent profile, awareness of site significance, awareness of environmental threats, attitudes toward responsibility and management, and reported behavioral practices such as waste disposal, compliance with site rules, and support for conservation measures. Most attitudinal items use a five-point Likert scale, while a smaller number of factual and behavioral items use categorical response formats.

A pilot survey is conducted before full deployment to test clarity, contextual relevance, and internal consistency. Necessary revisions are made after the pilot phase. Depending on field conditions, the final questionnaire may be administered through face-to-face survey, interviewer-assisted schedule, or mobile form-based entry.

3.5.3 Semi-Structured Interviews

Semi-structured interviews are conducted to explore the institutional and managerial dimensions of the study. Interview themes include perceived environmental threats, current management practices, enforcement challenges, inter-agency coordination, visitor management, awareness-building efforts, and policy implementation barriers. The semi-structured format allows the researcher to maintain comparability across interviews while also capturing detailed, context-specific insights.

3.5.4 Policy and Document Review

A systematic review of policy and institutional documents is undertaken to identify the formal framework governing heritage protection and environmental management at the site. The review includes UNESCO guidance relevant to World Heritage management, site-related documentation, and applicable national and local provisions concerning archaeology, environment, tourism, and site administration. The aim is to compare stated policy commitments with actual field practice.

3.6 Operationalization of Variables

Environmental pollution is operationalized as a composite of visible and site-relevant stress indicators, including litter, plastic waste, stagnant water, drainage-related vulnerability, vegetation growth, erosion, and unmanaged visitor impact. Stakeholder awareness is operationalized through knowledge of site significance, understanding of environmental threats, and familiarity with site rules. Environmentally responsible behavior is measured through self-

reported practices such as proper disposal of waste, compliance with movement restrictions, and support for conservation regulations. Policy implementation is operationalized through indicators such as monitoring frequency, signage adequacy, enforcement consistency, maintenance systems, and coordination among responsible agencies.

3.7 Data Analysis

Quantitative survey data are entered into statistical software and analyzed through descriptive statistics, including frequencies, means, and standard deviations. Inferential analysis is then used to test relationships among variables. Cross-tabulation and chi-square tests are used to examine group differences, while correlation and regression models are used where sample size and distribution permit.

Qualitative interview data are analyzed using thematic analysis. Interview transcripts and notes are coded iteratively to identify recurring themes such as monitoring challenges, coordination gaps, visitor pressure, awareness deficits, maintenance limitations, and institutional constraints. Spatial observation data are analyzed through zone-based comparison and, where possible, geospatial visualization. Pollution hotspot maps, circulation overlays, and management-priority maps are generated to identify concentrations of environmental stress. The final analytical step integrates quantitative, qualitative, and spatial findings to produce a multi-layered interpretation of site governance.

3.8 Reliability, Validity, and Ethics

The questionnaire is pilot-tested to improve reliability and internal consistency. Observation categories are standardized to reduce recording bias. Triangulation is built into the design by comparing findings from field observation, survey responses, interviews, and documentary review. Participants are informed of the purpose of the study, and confidentiality and voluntary participation are maintained throughout the research process.

4. Results Framework

4.1 Overview of the Results Structure

Because this study integrates spatial, social, and institutional data, the results section is organized around three analytical layers: environmental condition, stakeholder awareness, and policy implementation. This structure reflects the study's central premise that environmental stress at Somapura Mahavihara should be interpreted not as an isolated physical problem but as the outcome of interaction among site conditions, user practices, and governance capacity. In a completed empirical paper, this section should present observed field evidence first, followed by stakeholder survey findings, and then institutional and policy evidence. Such sequencing

allows the argument to move from the material condition of the site to the social and administrative processes that shape that condition.

4.2 Spatial Distribution of Environmental Pollution

The first results subsection should present the mapped distribution of environmental stress across the site. The analysis should identify which zones show higher concentrations of solid waste, plastic debris, stagnant water, drainage-related vulnerability, vegetation growth, surface degradation, erosion, or unmanaged circulation. In publication form, this subsection would ideally include a site map showing zone boundaries, a pollution intensity map, and a short table summarizing observation scores by zone.

The analytical purpose of this subsection is to distinguish between generalized environmental pressure and spatially concentrated risk. If the results show that environmental stress is not evenly distributed, this would support the proposition that pollution is shaped by local conditions such as visitor movement, edge neglect, water accumulation, or differential maintenance. Particular attention should be given to whether the central monument precinct, peripheral archaeological remains, circulation routes, or drainage-prone areas exhibit different levels of vulnerability. Such findings would align with conservation literature emphasizing the importance of preventive and spatially targeted management in archaeological landscapes (Feilden & Jokilehto, 1998; UNESCO et al., 2013).

4.3 Types of Pollution and Environmental Stress

A second results subsection should classify the major forms of environmental stress documented during fieldwork. These may include solid waste accumulation, plastic litter, stagnant or standing water, drainage obstruction, uncontrolled vegetation, soil erosion, visitor-related trampling, and visual disturbances affecting the site setting. The value of such categorization lies in showing that “pollution” in a heritage landscape is broader than waste alone. In an archaeological context, environmental harm includes processes that may not always be interpreted as pollution in a conventional urban sense but nonetheless threaten the integrity of the site.

If field evidence reveals the coexistence of maintenance-related problems and visitor-related impacts, the results should distinguish between them analytically. Maintenance-related stress would include drainage failure, unmanaged vegetation, or inadequate site servicing, whereas visitor-related stress would include littering, path deviation, and pressure on fragile areas. This distinction is important because it helps locate responsibility across different actors and institutions. It also strengthens the argument that site condition emerges from both behavioral and managerial causes.

4.4 Stakeholder Profile

Before presenting awareness data, the results section should briefly describe the composition of the respondent sample. This subsection should summarize respondents by stakeholder category, gender, age group, education level, and relationship to the site, such as visitor, resident, vendor, guide, site staff, or official. Such descriptive information is necessary because stakeholder awareness is unlikely to be uniform across groups. Previous literature has shown that social position influences knowledge, attitudes, and responsibility in heritage settings (Aas et al., 2005; Smith, 2006).

In the final article, a concise table may be used here to present demographic and stakeholder-group frequencies. This provides the interpretive basis for later comparisons in awareness and behavior.

4.5 Awareness of Heritage Value and Environmental Risk

This subsection should present survey findings relating to stakeholder awareness of Somapura Mahavihara's heritage significance, site fragility, environmental threats, and applicable rules or restrictions. In analytical terms, the results should distinguish among three dimensions of awareness: heritage awareness, environmental awareness, and regulatory awareness. Heritage awareness refers to understanding the cultural and historical value of the site. Environmental awareness refers to recognition of how pollution, water, vegetation, or improper behavior may damage the site. Regulatory awareness refers to familiarity with rules governing movement, waste disposal, or site protection.

If the data reveal that respondents appreciate the symbolic value of the site but remain less informed about practical conservation threats, this would be an important finding. Such a result would suggest that general respect for heritage does not automatically translate into informed stewardship. Similarly, if awareness differs substantially across visitors, residents, vendors, and officials, the results should highlight these differences through mean comparisons or cross-tabulations. Such variation would support the idea that management strategies must be stakeholder-specific rather than uniform.

4.6 Stakeholder Attitudes and Self-Reported Practices

This subsection should examine how respondents evaluate responsibility for site protection, whether they support stricter environmental regulation, and how they report their own behavior regarding waste disposal, compliance, and conservation-minded conduct. Where possible, the analysis should compare awareness and behavior scores to test the proposition that knowledge is associated with responsible practice. Such analysis would directly address the study's first hypothesis.

If higher awareness scores correspond to stronger support for environmental regulation or better self-reported behavior, this would be consistent with studies emphasizing the role of

public understanding in participatory heritage management (Aas et al., 2005; Smith, 2006). Conversely, if awareness is relatively high but environmentally responsible behavior remains weak, the results would suggest an implementation or behavior-enforcement gap. In either case, the section should be written to show whether stakeholder awareness functions as an enabling factor, an insufficient factor, or a weakly translated factor in site protection.

4.7 Institutional Findings and Policy Implementation Gaps

The next subsection should present evidence from interviews and document review concerning management practice, monitoring routines, maintenance constraints, inter-agency coordination, signage, public communication, staffing, and enforcement. Rather than merely listing institutional problems, the analysis should connect them to the environmental patterns already documented spatially. For example, if poorly monitored zones correspond to higher pollution or vegetation-related stress, the results should explicitly show that relationship.

Qualitative themes may include limited staff capacity, fragmented responsibility, weak operationalization of management plans, insufficient stakeholder communication, or the absence of routine environmental audits. These themes should be interpreted in relation to implementation theory, which emphasizes the gap that can arise between formal commitment and operational execution (Hill & Hupe, 2002; Pressman & Wildavsky, 1973). The key analytical goal here is to demonstrate whether policy exists largely at the normative level while implementation remains partial, inconsistent, or reactive.

4.8 Integrated Results

The final subsection of the results framework should integrate the spatial, social, and institutional findings. This is where the study's mixed-method design yields its strongest contribution. If environmental hotspots overlap with high-use areas, low-awareness groups, or weakly monitored zones, such overlap should be highlighted. If interviews indicate that certain recurrent problems are recognized by authorities but inadequately addressed due to resource or coordination limitations, that finding should also be incorporated.

The integrated results section should therefore answer the study's central question: how do environmental stress, stakeholder awareness, and policy implementation interact at Somapura Mahavihara? Rather than presenting the three dimensions in isolation, this subsection should show their interdependence. That is the point at which the case study moves beyond description and becomes an explanation of heritage vulnerability.

5. Discussion Framework

5.1 Interpreting Environmental Stress as a Heritage Governance Issue

The discussion should begin by reiterating that environmental pollution at Somapura Mahavihara is not simply a maintenance deficiency but a heritage governance issue. If the results show that pollution is concentrated in specific zones, shaped by circulation patterns, or intensified by weak monitoring, then environmental stress should be interpreted as a product of how space is used and managed. This interpretation is consistent with the broader heritage literature, which argues that the sustainability of cultural sites depends on integrated management rather than isolated technical intervention (Labadi, 2013; UNESCO et al., 2013).

In this context, the site's open archaeological character is especially significant. Unlike enclosed museum environments, exposed archaeological landscapes require continuous environmental stewardship, visitor guidance, and institutional responsiveness. The discussion should therefore argue that environmental pollution at such sites is structurally linked to accessibility, exposure, and governance capacity.

5.2 Stakeholder Awareness and Conservation Behavior

The discussion should then assess whether the study supports the proposition that awareness matters for environmentally responsible behavior. If the results show a positive association between awareness and conservation-supportive practices, that would confirm the importance of education and interpretation in site management. Such a finding would align with participatory heritage scholarship emphasizing that informed stakeholders are more likely to support protective measures and comply with rules (Aas et al., 2005; Smith, 2006).

However, the discussion should also consider the possibility that awareness alone is insufficient. If respondents demonstrate respect for the site but weak adherence to environmental behavior, this would suggest that awareness must be reinforced by visible regulation, infrastructure, incentives, and monitoring. The discussion should therefore distinguish between symbolic recognition of heritage and operational understanding of how to protect it. This distinction is especially important in publicly accessible heritage sites where appreciation does not automatically translate into compliance.

5.3 Policy Implementation and Institutional Weakness

A core contribution of the paper lies in connecting site condition to policy implementation. If interviews and document review reveal that environmental issues are recognized institutionally but not addressed systematically, then the findings would support implementation theory's claim that policy effectiveness depends on capacity, coordination, and routinization rather than formal declaration alone (Hill & Hupe, 2002; Sabatier & Mazmanian, 1980). The discussion should stress that implementation gaps do not necessarily indicate the absence of policy. Rather, they may reflect weak institutionalization of policy in daily management practice.

This is particularly relevant where environmental responsibilities are diffused across multiple agencies or where site staff operate under resource constraints. In such contexts, pollution and deterioration may persist not because they are unknown, but because they are insufficiently embedded in operational priorities. The discussion should therefore emphasize that policy implementation at heritage sites must include routine environmental audit, clearly assigned responsibilities, active visitor management, and mechanisms for feedback and accountability.

5.4 Spatial Management and Preventive Conservation

The mapped results should be used to argue for a shift from general maintenance to spatially differentiated preventive conservation. If certain areas of Somapura Mahavihara are shown to be consistently more vulnerable, the discussion should recommend zone-based management rather than uniform site-wide response. This would be consistent with conservation guidance that emphasizes prioritization, monitoring, and site-specific management planning (Feilden & Jokilehto, 1998; UNESCO et al., 2013).

Preventive conservation in this context would involve not only repairing damage after it occurs but also identifying where water accumulates, where vegetation is recurrent, where visitors deviate from controlled paths, and where archaeological remains are less visible but more exposed. The discussion should highlight that environmental mapping can function as a management tool rather than merely a research output.

5.5 Implications for World Heritage Management

The implications of the study extend beyond Somapura Mahavihara. Many archaeological World Heritage Sites in South Asia and elsewhere face similar conditions: exposed fabric, open circulation, seasonal environmental stress, tourism pressure, and uneven implementation of conservation policy. The discussion should therefore argue that the integrated framework used in this study has transferable value. By linking environmental observation, stakeholder analysis, and policy review, the paper offers a practical model for other heritage sites seeking to move beyond monument-centered preservation toward landscape-based and participatory management.

At the same time, the discussion should avoid overgeneralization. Somapura Mahavihara has its own historical, spatial, and administrative context. The paper's broader contribution is methodological and conceptual rather than universally predictive. It shows how environmental degradation can be analyzed not only as a physical risk but as a governance process.

6. Policy Implications and Management Recommendations

The findings of this study should culminate in a set of policy and management implications grounded in the evidence. First, site management should adopt a **zone-based environmental monitoring system** in which vulnerable areas are regularly inspected and documented. Second, **routine pollution audits** should be institutionalized, including mapping of solid waste, drainage issues, vegetation growth, and visitor-pressure zones. Third, **stakeholder awareness programs** should be tailored to different user groups rather than delivered generically. Visitors may need concise interpretive messaging, while local actors may benefit from repeated community-based engagement.

Fourth, **signage, circulation guidance, and waste infrastructure** should be improved where behavioral issues are concentrated. Fifth, **institutional coordination mechanisms** should be clarified so that environmental management responsibilities do not remain fragmented across agencies. Sixth, **heritage management plans** should include explicit environmental indicators, implementation timelines, and review procedures. These recommendations align with broader international calls for integrated, participatory, and preventive heritage management (UNESCO et al., 2013).

7. Conclusion

This paper has argued that environmental pollution at Somapura Mahavihara should be understood not merely as a problem of site maintenance but as a multidimensional issue shaped by environmental condition, stakeholder awareness, and policy implementation. By positioning pollution within a broader heritage governance framework, the study moves beyond descriptive concern about deterioration and instead examines how archaeological vulnerability is produced through the interaction of material exposure, human behavior, and institutional practice.

The study contributes conceptually by linking archaeological conservation to environmental governance and implementation analysis. It contributes methodologically by combining spatial pollution mapping, stakeholder survey, interviews, and document review within a mixed-method case study design. It contributes practically by offering a structure through which site managers can identify environmental hotspots, detect awareness gaps, and evaluate whether policy commitments are being operationalized in everyday management. In doing so, it responds to longstanding calls in heritage scholarship for more integrated and participatory approaches to conservation (Aas et al., 2005; Smith, 2006; UNESCO et al., 2013).

The central implication of the study is that effective protection of Somapura Mahavihara requires more than technical conservation of exposed brick remains. It requires a management system in which environmental monitoring is routine, stakeholder awareness is cultivated, and

policy implementation is made visible through clear responsibilities, preventive action, and institutional coordination. Such an approach is likely to be relevant not only for Somapura Mahavihara but also for other archaeological heritage landscapes where environmental pressure and governance fragility intersect. Future empirical work based on full field data can test the hypotheses proposed here more directly and refine the policy model for broader comparative application.

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